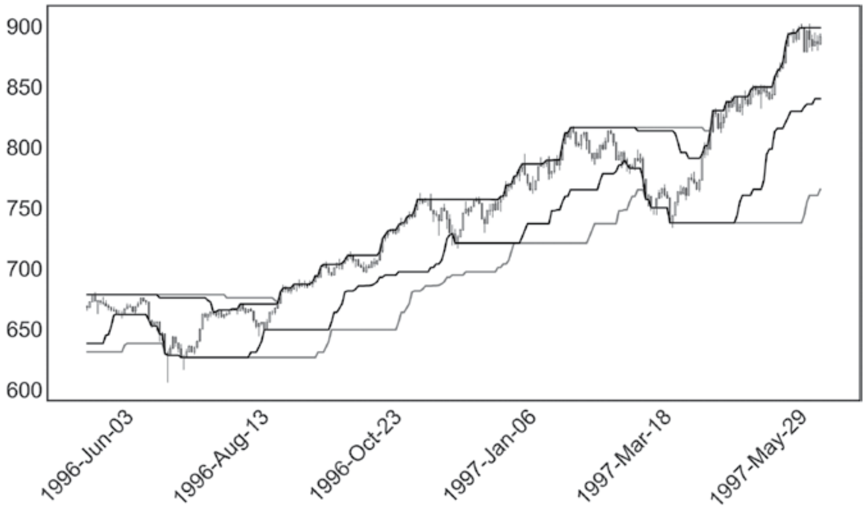

FIGURE 4.7 **Breakout strategy without trend filter.**

Figure 4.7 shows the breakout strategy in a strongly trending period of the S&P 500 at the height of the dot com bubble. Two things should be clear just from a quick glance: the strategy enters into shorts in a strong bull market, which does not make sense, and it over-trades, taking too many positions back and forth in a strong trending market.

The remedy for both these situations is to add a trend filter, which makes sure that we only trade in the direction of the main trend and that we don't get whipsawed by going in and out of positions every couple of days. The moving average strategy does in fact make for a pretty good trend filter by itself, so let's try to use it just the way it is. We're going to simply combine the two basic strategies into one, by trading the breakout strategy, using the moving average strategy as a trend filter. There is nothing really special about the parameters and details of either of these strategies and the numbers used are deliberately kept round and arbitrary. No optimisation or other shenanigans. The concept is what matters rather than the exact details.

Figure 4.8 shows crude oil during a bull market phase followed by a bear market phase. The method used to determine which market regime we are in is a combination of two exponential moving average lines, where the solid line is the slower 100-days moving average and the dotted line is the faster-reacting 50 days. Here the moving average lines are not used for trading signals because it was in one of our initial strategies, but just as a filter to see